

FORMAL RESEARCH PROPOSAL & TECHNICAL SPECIFICATION

Quantifying AI Over-Reliance in Programming Education: A Telemetry-Driven Behavioral Metric with Productive Friction in S-SPARC

A Complete Empirical Framework for Diagnosing Epistemic Dependence, Latent Construct Modeling, and Closed-Loop Scaffolding on the E-STRANGE Learning Platform

PRINCIPAL INVESTIGATOR	RESEARCH DOMAINS	OPERATIONAL TESTBED	TARGET PUBLICATION
Faculty Research Team • S-SPARC Group	AIED, EDM, Learning Analytics	S-SPARC Engine & E-STRANGE LMS	IEEE TALE / Scopus Q1 (IEEE TLT)

EXECUTIVE SUMMARY

Generative AI accelerates software development but introduces severe pedagogical risks in computer science education, including cognitive offloading, epistemic dependence, and degradation of analytical self-debugging skills. This research proposal details the design, mathematical formulation, data architecture, and empirical validation of the **AI Over-Reliance Index (ORI)**. Integrated within the S-SPARC AI assistant and E-STRANGE automated assessment system, the framework captures millisecond-level telemetry (Shannon entropy, C-I-O-E problem completeness, conceptual ratio, keystroke cadence) with zero manual survey overhead. We establish a robust two-phase psychometric roadmap (Phase 1: Exploratory Factor Analysis; Phase 2: Independent Confirmatory Factor Analysis and convergent validation against the standardized Hou et al. 2025 scale) while resolving critical methodological threats including maturation confounders, entropy inversion paradox, classifier error propagation, and Pygmalion bias.

1. Introduction & Problem Formulation

The proliferation of Large Language Models (LLMs) has introduced a critical challenge to higher computing education: while students can generate syntactically correct code instantaneously, unassisted problem-solving ability declines. In laboratory sessions, students frequently execute a passive feedback loop:

Copy Assignment Prompt → Paste into AI → Copy Output Code → Submit to Auto-Grader

Because automated graders evaluate only input/output correctness, students pass unit tests without engaging in syntax tracing, logic decomposition, or algorithmic reflection. This phenomenon (cognitive atrophy) produces severe failures during closed-book, unassisted evaluations [2], [3], [6].

2. State of the Art & Research Gap

Methodological Paradigm	Critical Limitations & Vulnerabilities	S-SPARC Telemetry Framework
Subjective Likert Surveys [2], [7]	Recall bias, social desirability, cannot measure live behavior during coding.	Passive Millisecond Telemetry: Automatic logging of typing speed, pauses, and prompt quality per millisecond.
Binary Plagiarism Detectors	High false positives, purely punitive, zero formative learning guidance.	Cognitive Inquiry Profiling: Longitudinally maps problem decomposition and epistemic autonomy [5].
Post-Hoc Eye-Tracking [16]	High laboratory rigor but cannot run continuously in active classroom settings.	Real-Time Metric (ORI): Live composite calculation triggering dynamic adaptive scaffolding.
Unregulated AI Tools [5]	Encourages prompt spamming and full code outsourcing without reflection.	Productive Friction: Mandatory 60s pause with stepped Socratic hints [9], [10].

3. Mathematical Metric Formulation (AI Over-Reliance Index)

The composite AI Over-Reliance Index (ORI) is defined on the domain [0, 1], where 0.0 represents fully autonomous problem-solving and 1.0 indicates complete epistemic dependence:

$$ORI = 1 - (w1 \cdot Q_prompt + w2 \cdot S_CIOE + w3 \cdot R_concept + w4 \cdot F_reflect)$$

Where indicators are normalized to [0, 1] • Baseline exploratory weights: w1=0.30, w2=0.25, w3=0.20, w4=0.25

1. Shannon Prompt Entropy (Q_prompt)

$H(X) = -\sum p(x) \log_2 p(x)$ [11]. Measures lexical diversity and syntax token density in prompts [13], [17]. Penalizes vague queries like "fix my code". Baseline weight: w1 = 0.30.

2. Context Completeness (S_CIOE)

Evaluates 4 structural dimensions: Context, Input specs, Expected Output, and Error Trace [12]. Rewards rigorous problem framing. Baseline weight: w2 = 0.25.

3. Conceptual Ratio (R_concept)

Ratio of conceptual inquiries (Bloom C2-C4) versus direct code generation requests (Bloom C1). Differentiates learning from outsourcing. Baseline weight: w3 = 0.20.

4. Friction Adherence (F_reflect)

Tracks adherence to the 60s reflection pause [9] with keystroke cadence verification to detect external copy-pasting. Baseline weight: w4 = 0.25.

4. Relational Database Telemetry Schema

The data collection pipeline operates passively within the MySQL database of S-SPARC and E-STRANGE:

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Database Variable	Data Definition & Measurement Granularity	Construct Mapping
<code>prompt_raw_text</code>	UTF-8 query string with millisecond timestamp	Query formulation quality
<code>entropy_score</code>	Shannon Information Entropy $H(X)$ calculated over token stream	Lexical richness (Q_{prompt})
<code>cioe_bitmask</code>	4-bit vector for Context, Input, Output, and Error stack trace	Problem completeness (S_{CIOE})
<code>keystroke_dwell</code>	Inter-keystroke timing variance and anti-paste dwell metrics	Friction integrity (F_{reflect})
<code>execution_metrics</code>	Unit test pass rate, runtime errors, and compilation attempts	Objective mastery
<code>carbon_footprint</code>	Estimated inference token energy consumption and CO2 (g)	Compute sustainability

5. Four Secondary Exploratory Hypotheses (Bonferroni Corrected $\alpha = 0.0125$)

H1: Green AI Volume Paradox

Over-reliant students generate higher cumulative carbon emissions due to excessive prompt iteration volumes rather than per-query token complexity (controlled for Module Fixed-Effects and semantic cache hits).

H2: AST Structural Distance Delta

Autonomous students exhibit significantly larger Abstract Syntax Tree (AST) structural distances between AI suggestions and final submitted code (evaluated against the final session AI suggestion).

H3: Reflexive Pause vs First-Attempt Pass Rate

Adherence to reflection pauses positively predicts first-attempt pass rates on programming tasks (controlling for baseline prior ability as a covariate; claims remain strictly associative).

H4: Semantic Cache Novelty & Lifecycle

Semantic cache misses reflect deeper student exploratory inquiry into complex edge cases (controlled for cache maturity lifecycle to eliminate cold-start artifacts).

6. Methodological Defenses & Threats to Validity

- **Reactivity & Maturation Confound Mitigation:** The study employs an A-B design (Modules 1-2 baseline unconstrained vs Modules 3-4 active 60s friction). For a single-cohort setting, the maturation effect of time is formally disclosed as a limitation; if 2 parallel sections are available, counterbalancing (A-B vs B-A) is implemented.
- **Two-Phase Psychometric Roadmap:**
 - **Phase 1 (Semester 1 / Cohort 1, N=35-50):** Instrument development and Exploratory Factor Analysis (EFA) to discover latent factor structure and calibrate initial weights.
 - **Phase 2 (Semester 2 / Cohort 2, N ≥ 80):** Independent Confirmatory Factor Analysis (CFA), construct validation against the standardized Hou et al. (2025 [1]) scale, and testing interaction models without factor overfitting.
- **Entropy Inversion Paradox (Interaction Modeling):** Advanced students author brief, highly specific prompts with low raw entropy. We model an interaction term ($\text{Entropy} \times \text{Baseline Prior Ability}$) so expert conciseness is not misdiagnosed as novice over-reliance.
- **NLP Classifier Validation Sub-Study:** Intent classification for `R_concept` is validated on ≥ 100 sample prompts with double annotation by 2 independent raters, requiring Cohen's Kappa $\kappa \geq 0.80$.
- **Ethical Masking & Pygmalion Effect:** Individual student ORI scores and personas are blind-masked from instructors during Phase 1 to prevent subjective grading bias. Live lecturer dashboards unlock in Phase 2 post-validation.

- **Anti-Circularity & P-Hacking:** The analysis plan is pre-registered on OSF.io. Regresi bobot diuji dengan 5-Fold Cross-Validation, and VIF is checked ($VIF < 5.0$).

7. Implementation Timeline & Two-Phase Milestones

Period	Research Activities	Key Deliverables
Weeks 1 - 2	OSF Pre-Registration, Ethics Approval, Baseline Phase (Modules 1-2)	Locked OSF protocol, baseline interaction logs
Weeks 3 - 4	Intervention Phase (Modules 3-4 with active 60s productive friction)	Complete pre/post telemetry dataset
Weeks 5 - 6	MySQL ETL extraction, NLP double annotation ($\kappa \geq 0.80$), EFA factor structure	Psychometric factor report
Weeks 7 - 8	Manuscript drafting and submission to IEEE International Conference	Submitted paper (IEEE TALE / EDUCON)
Semester 2	Phase 2 independent replication (CFA, Hou et al. validation, Scopus Q1 submission)	Scopus Q1 Journal (IEEE TLT / C&E: AI)

8. Scope Boundary & Advisor Collaboration

Methodological Scope Boundary The ORI metric is specifically formulated for programming assignments evaluable via deterministic unit testing (Computer Science). Generalization to non-computing fields requires qualitative rubrics and is defined as future work.	Advisor Action Items 1. Approval to deploy S-SPARC in 1 active lab class (35-50 students). 2. Ethics review endorsement and OSF co-sponsorship. 3. Collaborative statistical review and co-authorship through publication.
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9. Academic References (IEEE Citation Standard)

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